Mahler series with multiplicative coefficients

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Two kinds of structure

Our motivating goal is to understand structures that are well-behaved with respect to two different notions. For this talk, the relevant notions for a sequence will be:

- (computational) The n-th term can be computed via the base-k digits in some nice way (e.g., k-automatic).
- (arithmetic) Its values of the n-th term be understood in terms of the prime factorization of n (multiplicative sequences).

The main question

How much freedom remains when both descriptions apply?

k-automatic sequences

Obligatory example: Let $t(n)$ be the parity of the number of 1's in the binary expansion of n.

$$(t(n))_{n\geq 0} = 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, \ldots$$

Only two states are needed: even and odd.

$$t(2n) = t(n), \qquad t(2n+1) = 1 - t(n).$$

A sequence is k-automatic if a finite automaton, with an output at each state, computes it from the base-k digits of n.

Cobham's theorem

Cobham (1969)

If k and ℓ are multiplicatively independent, a sequence is both k-automatic and ℓ-automatic if and only if it is eventually periodic.

$$\text{Multiplicatively independent}: \quad k^a \neq \ell^b \qquad \text{for all } a, b \geq 1.$$

For example, base 2 and base 3 are independent.

Base k^i and base k^j describe the same automatic sequences, so we need this constraint.

A. Cobham, *Math. Systems Theory* 3 (1969), 186–192.

The arithmetic condition

We call $f : \mathbb{N} \to K$ multiplicative if

$$f(mn) = f(m)f(n) \qquad \text{whenever } \gcd(m, n) = 1.$$

For a nonzero multiplicative function, $f(1) = 1$, and

$$f(n) = \prod_{p^a \| n} f(p^a).$$

Examples include $\mu(n)$ (the Möbius function) and $\varphi(n)$ (the Euler φ function).

Complete multiplicativity asks for the identity for all m, n.

There really are nonperiodic examples

Consider

$$a(n) = (-1)^{v_2(n)}, \qquad n \geq 1.$$

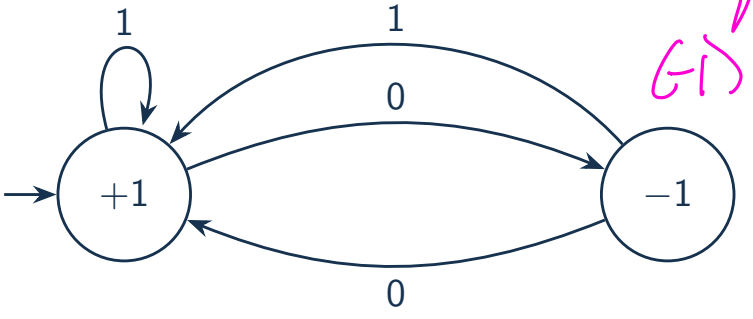
n	1	2	3	4	5	6	7	8	9	10	11	12
$a(n)$	1	-1	1	1	1	-1	1	-1	1	-1	1	1

It is completely multiplicative, since valuations add.

$$a(2n) = -a(n), \qquad a(2n+1) = 1.$$

So it is also 2-automatic.

The automaton sees the final zeros



Read the standard binary expansion from left to right.

The output records the parity of the number of trailing zeros.

From nonautomaticity to classification

- **Yazdani (2001):** nonautomaticity results for reductions of familiar multiplicative functions modulo integers.
- **Coons (2010):** nonautomaticity of the Liouville function and further number-theoretic examples.

The broader question becomes:

Which multiplicative functions can be automatic at all?

Yazdani, JTNB 13 (2001), 651–658, doi:10.5802/jtnb.342.

Coons, JTNB 22 (2010), 339–352, doi:10.5802/jtnb.718.

First understand the large primes

A conjecture of B–Bruin–Coons (2012)

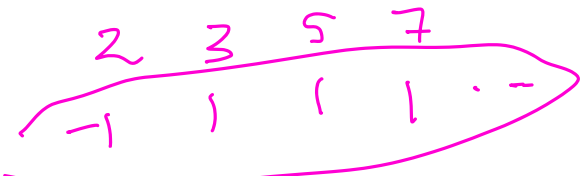
If f is multiplicative and automatic, its values on primes agree with an eventually periodic function.

Konieczny and, independently, Klurman–Kurlberg proved more:

On integers $n > 1$ with no small prime factors, f agrees with a Dirichlet character, or is identically zero.

Konieczny, *Bull. Lond. Math. Soc.* (2020), doi:10.1112/blms.12317.

Klurman–Kurlberg, ibid. (2020), doi:10.1112/blms.12318.



The automatic classification

Konieczny–Lemańczyk–Müllner (2022)

Every multiplicative automatic sequence is p-automatic for some prime p, and has the form

$$f(p^i m) = g(i)\chi(m) \qquad (p \nmid m).$$

Here g is eventually periodic, $g(0) = 1$, and χ is multiplicative and eventually periodic.

Conversely, every such formula gives a multiplicative p-automatic sequence.

Math. Z. 300 (2022), 1297–1318. doi:10.1007/s00209-021-02834-3.

Why the base matters

$k = p^e$

k-automatic means p-automatic.

The exponent $v_p(n)$ can carry an eventually periodic sequence.

k has at least two distinct prime factors

The classification also makes f p-automatic for some prime p.

k and p are independent.

Cobham forces eventual periodicity.

A second classification to keep in mind

Bézivin; Bell–Bruin–Coons

Over a field of characteristic zero, if

$$F(x) = \sum_{n \geq 1} f(n) x^n$$

is algebraic and f is multiplicative, then

$$f(n) = n^r \chi(n), \qquad r \in \mathbb{Z}_{\geq 0},$$

with χ multiplicative and eventually periodic.

In particular, F is rational.

Bézivin (1995); Bell–Bruin–Coons, *Trans. AMS* 364 (2012), 933–959.

Finite output is a serious restriction

Automatic sequences take only finitely many values.

But natural multiplicative examples include

$$f(n) = n, \qquad f(n) = 3^{v_3(n)}, \qquad f(n) = \frac{n}{2^{v_2(n)}}.$$

All three admit finite-dimensional descriptions from binary digits. Can we replace automaticity by something more general?

The kernel of a sequence

The k-kernel of f is

$$\mathcal{K}_k(f(n)) = \left\{ \left(f(k^e n + a)\right)_{n \geq 0} : e \geq 0,\ 0 \leq a < k^e \right\}.$$

These are all the subsequences obtained by repeatedly fixing the last digit.

For Thue–Morse,

$$\mathcal{K}_2(t(n)) = \{t(n), 1 - t(n)\}.$$

Equivalent definition

f is k-automatic if and only if $\mathcal{K}_k(f)$ is finite.

Regular sequences

Allouche–Shallit

A K-valued sequence f is k-regular if

$$\dim_K \operatorname{span}_K \mathcal{K}_k(f) < \infty.$$

For example, if $f(n) = n$, then

$$f(k^e n + a) = k^e n + a,$$

so the kernel lies in the span of $n \mapsto n$ and $n \mapsto 1$.

Every automatic sequence is regular. A finitely valued regular sequence is automatic.

Allouche–Shallit, *Theoret. Comput. Sci.* 98 (1992), 163–197.

An unbounded example: the odd part

Put

$$b(n) = \frac{n}{2^{v_2(n)}} \quad (n \geq 1), \qquad b(0) = 0.$$

Then

$$b(2n) = b(n), \qquad b(2n+1) = 2n+1.$$

The space spanned by $b(n)$, n, and 1 is stable under taking even and odd subsequences.

Thus b is 2-regular and multiplicative, but not automatic.

Regular sequences: Digits now act by matrices

A k-regular sequence has a representation

$$f(n) = uA(d_s)\cdots A(d_0)v, \qquad n = (d_s\cdots d_0)_k,$$

with fixed finite matrices $A(0),\ldots,A(k-1)$, and row vector u and column vector v.

In particular, for a prime base p,

$$f(p^i) = uA(1)A(0)^i v.$$

A useful consequence

The sequence $i \mapsto f(p^i)$ satisfies a constant-coefficient linear recurrence.

A further extension: Mahler equations

A series $F(x) \in K[[x]]$ is k-Mahler if

$$P_0(x)F(x) = \sum_{j=1}^{d} P_j(x)F(x^{k^j}), \qquad P_j \in K[x], \quad P_0 \neq 0.$$

Inhomogeneous equations give the same class.

For the odd-part series $B(x) = \sum_{n\geq 1} b(n)x^n$,

$$B(x) = B(x^2) + \frac{x(1+x^2)}{(1-x^2)^2}.$$

automatic $\subseteq$ regular $\subseteq$ Mahler.

Mahler series can be much less controlled

Consider the formal product

$$E(x) = \prod_{j \geq 0} \frac{1}{1 - 2x^{2^j}}.$$

It satisfies

$$(1 - 2x)E(x) = E(x^2).$$

2-Mahler

But its coefficients satisfy $[x^n]E(x) \geq 2^n$.

A complex-valued regular sequence has at most polynomial growth.

So E is Mahler but not regular.

Cobham's theorem reappears

Adamczewski–B (2017), Also Schäfke-Singer

If k and ℓ are multiplicatively independent, then

$$F \text{ is both } k\text{-Mahler and } \ell\text{-Mahler} \iff F \text{ is rational.}$$

For finite-valued coefficients, rationality reduces to eventual periodicity.

This theorem will also enter the proof of our result.

Ann. Sc. Norm. Super. Pisa 17 (2017), 1301–1355.

What should the answer look like?

We have two models:

$$\text{multiplicative} + \text{automatic} \quad : \quad f(p^i m) = g(i)\chi(m),$$

$$\text{multiplicative} + \text{rational} \quad : \quad f(n) = n^r \chi(n).$$

A natural candidate combines them:

$$f(p^i m) = g(i) m^r \chi(m).$$

Does allowing arbitrary Mahler equations introduce anything else?

The main reduction: regularity

B–Smertnig

Let K have characteristic zero, and let $k \geq 2$.

If

$$F(x) = \sum_{n \geq 1} f(n) x^n$$

is k-Mahler and f is multiplicative, then f is k-regular.

Multiplicativity rules out the extra freedom of general Mahler series.

B–Smertnig, arXiv:2603.23456, Theorem 1.3.

The classification in a prime-power base

Suppose $k = p^e$. Then

$$f(p^i m) = g(i) m^r \chi(m) \qquad (p \nmid m),$$

where

- $g(i)$ is a linear recurrence sequence, with $g(0) = 1$;
- r is a nonnegative integer;
- χ is multiplicative and eventually periodic.

Conversely, every such formula is multiplicative and p-regular.

B–Smertnig, Theorem 1.3 and Proposition 3.4.

The classification in every other base

B–Smertnig

If k has at least two distinct prime divisors and F is k-Mahler with multiplicative coefficients, then F is rational.

Consequently,

$$f(n) = n^r \chi(n)$$

with $r \geq 0$ and χ multiplicative and eventually periodic.

If a multiplicative Mahler series is nonrational, its exceptional prime is unique.

B–Smertnig, Corollary 1.4; uniqueness follows from Adamczewski–B.

The freedom at one prime

Write $n = p^i m$, with $p \nmid m$. Take $\chi = 1$.

$g(i)$	r	$f(n)$
$(-1)^i$	0	$(-1)^{v_p(n)}$
$i+1$	0	$1 + v_p(n)$
1	1	$n/p^{v_p(n)}$
F_{i+1}	0	$F_{v_p(n)+1}$

Here $F_0 = 0$, $F_1 = 1$ are the Fibonacci numbers.

The generating series behind the formula

Put

$$H(x) = \sum_{\substack{m \geq 1 \\ p \nmid m}} m^r \chi(m) x^m.$$

This is rational (the part of the series from n coprime to p). The classification gives

$$F(x) = \sum_{i \geq 0} g(i) H(x^{p^i}).$$

The part away from p is rational.

The recurrence for g controls how these transforms of H fit together.

A consequence for familiar functions

The generating series of $\varphi(n)$ is not k-Mahler for any $k \geq 2$.

For example, if φ had the classified form, then there would be a distinguished prime p and for every prime $q \neq p$,

$$q - 1 = \varphi(q) = q^r \chi(q).$$

Since χ takes finitely many values, this is impossible since $(q-1)/q^r$ takes infinitely many values.

The proof has two stages

1. **Classify the regular case.**

   Reduce to finite fields and use the automatic classification.

2. **Show that multiplicative Mahler series are regular.**

   Use multiplicativity to control the Mahler denominator.

We use the earlier classification theorems (both for rational multiplicative and automatic multiplicative sequences).

Finite fields turn regular into automatic

A regular sequence has a matrix representation over a finitely generated ring $R \subseteq K$.

For each maximal ideal $\mathfrak{m}$,

$$R/\mathfrak{m} \text{ is finite}, \qquad \overline{f} \text{ is regular and finitely valued.}$$

Thus $\overline{f}$ is automatic and multiplicative.

If k has two distinct prime factors, KLM and Cobham make every $\overline{f}$ eventually periodic.

A lifting theorem for regular sequences then makes F rational.

B (2005), Theorem 3.11; B–Smertnig, Proposition 3.1.

Remove the exceptional prime

Now suppose f is p-regular. Define

$$h(n) = \begin{cases} f(n) & p \nmid n, \\ 0 & p \mid n. \end{cases}$$

This is still regular and multiplicative.

After reduction to a finite field, the automatic classification makes h eventually periodic: its p-power part has been removed.

The same lifting theorem makes $H(x) = \sum h(n)x^n$ rational.

$$h(n) = n^r \chi(n).$$

B–Smertnig, Lemma 3.3, using the rational multiplicative classification.

The regular classification is now forced

Multiplicativity gives

$$f(p^i m) = f(p^i) f(m) \qquad (p \nmid m).$$

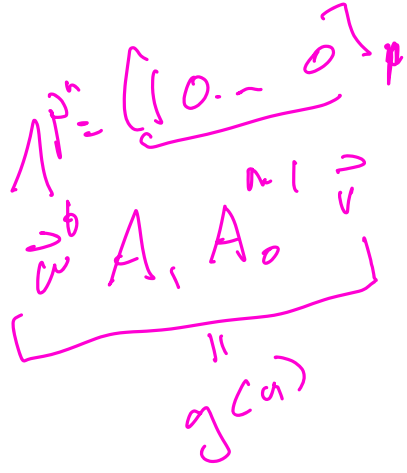
From the previous slide, $f(m) = m^r \chi(m)$ for $p \nmid m$.

From the matrix representation,

$$g(i) := f(p^i) = uA(1)A(0)^i v$$

is a linear recurrence sequence.

$$f(p^i m) = g(i) m^r \chi(m).$$



What obstructs regularity?

In a Mahler equation,

$$P_0(x)F(x) = \sum_{j \geq 1} P_j(x)F(x^{k^j}),$$

consider every possible leading coefficient P_0.

These polynomials form an ideal in $K[x]$. Its normalized generator is the Mahler denominator $\mathfrak{d}_k F$.

It divides every possible P_0, and itself occurs as a leading coefficient.

B–Smertnig, Definition 2.3.

A precise regularity criterion

Adamczewski–B–Smertnig (2023)

A k-Mahler series F is k-regular if and only if every nonzero root α of $\partial_k F$ is a root of unity whose order has a common factor with k.

For $k = 2$, even-order roots of unity are allowed.

The root 1, odd-order roots of unity, and non-torsion roots are obstructions.

We need to show that multiplicativity removes all these obstructions.

J. Eur. Math. Soc. 25 (2023), Theorem 10.4.

A series measuring failure of complete multiplicativity

For a large prime $q \nmid k$, let ω be a primitive qth root of unity. Set

$$D_q(x) = \sum_{j=0}^{q-1} F(\omega^j x) - qf(q)F(x^q).$$

The root-of-unity filter gives

$$D_q(x) = q \sum_{n \geq 1} (f(qn) - f(q)f(n)) x^{qn}.$$

Multiplicativity makes the summand zero when $q \nmid n$.

$$D_q(x) \in K[[x^{q^2}]].$$

Sparsity forces the defect to be regular

Two facts about the Mahler denominator interact:

- Since $D_q(x) \in K[[x^{q^2}]]$, its denominator has the form $Q(x^{q^2})$.
- Since D_q is built from $F(\omega^j x)$ and $F(x^q)$, its denominator roots are controlled by those of $\mathfrak{d}_k F$.

For sufficiently large q, a root outside the allowed set would force incompatible root-of-unity relations.

Key lemma

D_q is k-regular for every sufficiently large prime q.

B–Smertnig, Section 4 and Lemma 5.4.

Recovering f

Suppose

$$c = f(q^i) - f(q)f(q^{i-1}) \neq 0$$

for some $i \geq 2$. For every m with $q \nmid m$,

$$[x^{q^i m}]D_q(x) = qc\, f(m).$$

So regularity of D_q gives regularity of f restricted to $m \not\equiv 0 \pmod{q}$.

Gluing this information together for different q then gives regularity of F.

B–Smertnig, Proposition 5.1.

But what happens when this doesn't happen?

If we don't have this condition, we have $f(qn) = f(q)f(n)$ for all n.
Thus

$$\sum_{j=0}^{q-1} F(\omega^j x) = qf(q)F(x^q).$$

Iterating this identity first forces all nonzero denominator roots to be roots of unity.

We then work a bit to remove the remaining forbidden roots.

Mahler $\Rightarrow$ regular

B–Smertnig, Sections 6–7, especially Theorem 7.1.

The two types of structure revisited

For a multiplicative Mahler series:

- The argument above forces regularity.
- Finite-field reductions give a rational part away from one prime.
- Matrix powers give a linear recurrence along powers of that prime.

$$f(p^i m) = g(i) m^r \chi(m) \qquad (p \nmid m).$$

In a base with two distinct prime factors, that exceptional-prime freedom disappears.

A question about recognizing the structure

Suppose a Mahler equation over $\mathbb{Q}(x)$, together with enough initial data, specifies a unique series F.

Question

Can we decide whether its coefficients are multiplicative?

If so, can we recover p, r, a recurrence for g, and the eventual periodic data for χ?

The classification describes what a positive answer must look like. It does not, by itself, give a stopping rule for testing multiplicativity.

A discussion question suggested by this work; no claim of a new open problem, since AI (and perhaps even I) can probably solve this. This is intended to be just a starting point to look at decidability phenomena for this type for Mahler series.

What if multiplicativity is replaced by additivity?

Suppose instead that

$$a(mn) = a(m) + a(n) \quad \text{when } \gcd(m, n) = 1.$$

Natural p-regular examples are

$$a(n) = h(v_p(n)),$$

where h is a linear recurrence sequence with $h(0) = 0$.

Question

Which additive arithmetic functions have Mahler generating series? Is there a comparable description involving one exceptional prime?

A discussion question, not a claimed new open problem, because if AI solves it right away, then it's not *really* the right question.

Thanks!